PAPER_276_Andromeda_FriedmannUQFF_HzExpansionCoupling_H_UQFF_NearUnityResonance

Daniel T. Murphy

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PAPER_276: UQFF **Date:** March 2026 $n = [int]# UQFF PAPER_276 ## Andromeda Friedmann-UQFF Gravity Coupling: H(z)t Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy
Session: 76 (March 2026)
Module: ANDROMEDA_UQFF_MODULE.cpp (M31 Master, UQFF 2.0)
WOLFRAM_TERM: `AndromedaUQFF:g_exp=G`*`M/r^2`*`H(z)`*`t;`
`H(z)=H0``Sqrt[Om_m`*`(1+z)^3+Om_L]/Mpc2m;`*
`H_UQFF=``H(z)`*`t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g_expansion = (μ_s∇(M_s/r)) H(z) t. For Andromeda's blueshift redshift z
= -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = 2.269$\times$10?8 s-1. Evaluated at the Hubble timescale t = t_H = 4.352$\times$10-7 s, the dimensionless Friedmann-UQFF Resonance Coefficient H_UQFF =
H(z)$\cdot$t_H **0.987** a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M_visible/M_DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

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1. Background

The UQFF framework computes total gravitational acceleration g_total(r,t) as a sum of DPM-seeded, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms.

PAPER_273275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)t 1 implying that expansion-mediated gravitational coupling is of the same order as the Step 10 Newton observational projection term.

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2. Friedmann H(z) Expansion Coupling

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The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_Lambda}$$

For Andromeda parameters: H0 = 70.0 km/s/Mpc, O_m = 0.3, O_? = 0.7, z = -0.001.

$$H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: H_SI = 69.969 $\times$ 10 / 3.086$\times$10 = **2.269$\times$10?8 s-1**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda:
- g_base = μ_s∇(M_s/r) = 6.674$\times$10? 1.989$\times$104 / (1.04$\times$10) = **1.227$\times$10? m/s**
- At t = 1 Gyr (3.156$\times$10-6 s): g_expansion = 1.227$\times$10? 2.269$\times$10?8 $\times$ 3.156$\times$10-6 = **8.79$\times$10? m/s**
- At t = t_Hubble (4.352$\times$10-7 s): g_expansion = 1.227$\times$10? $\approx$ 0.987 = **1.211$\times$10? m/s**

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3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where t_H = 4.352$\times$10-7 s (Hubble time 13.8 Gyr).

For Andromeda (z = -0.001):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

H_UQFF $\approx$ 0.987 **1** is a remarkable result. It means:

> Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the Step 10 Newton observational projection term** nearly doubling g_base.

This near-unity value is not coincidental. In a flat ?CDM universe (O_m + O_? = 1):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because H0 t_H is the dimensionless Hubble number $\approx$ 0.96§1.0 for observationally consistent H0.

New UQFF Constant: H_UQFF = H(z)$\cdot$t_H **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269$\times$10?8 s-1
t_H	4.352$\times$10-7 s
H_UQFF	**0.987** (~1)
g_expansion(t_H)	1.211$\times$10? m/s
g_base	1.227$\times$10? m/s
Ratio g_exp/g_base	98.7%

3.3 Blueshift Sensitivity

For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O_m(1+z) is suppressed below O_m for z < 0), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

For Andromeda (z = -0.001): H_UQFF = 0.987 vs flat-universe H_UQFF0 = 0.9985. The blueshift suppresses H_UQFF by 0.15%.

This is the inverse of the redshift behaviour (z > 0 ? H(z) > H0 for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

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A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where ?_mean = M/V_fluid = 1.989$\times$104 / 106 = 1.989$\times$10?8 kg/m.

Numerical value for Andromeda:
$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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5. M_visible / M_DM_mass Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$
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For Andromeda defaults: M = 1.989$\times$104 kg, f_DM = 0.80:
- M_visible = 0.20 $\times$ 1.989$\times$104 = **3.978$\times$104 kg** (20% visible baryons)
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These are tracked in `updateCache()` and exported via `exportState()` (params 2632), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

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Term magnitudes at t=0 (current epoch):

Term	Value (m/s)	Paper
g_grav = μ_s∇(M_s/r)	1.227$\times$10?	baseline
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Seven new parameters added to `exportState()`:
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$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

---

5. M_visible / M_DM_mass Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$
$$M_{\text{DM,mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: M = 1.989\times104 kg, f_DM = 0.80:
- M_visible = 0.20 \times 1.989\times104 = **3.978\times104 kg** (20% visible baryons)
- M_DM_mass = 0.80 \times 1.989\times104 = **1.591\times104 kg** (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 2632), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

---

6. Updated ANDROMEDA_UQFF_MODULE.cpp g_total

The full UQFF 2.0 equation for Andromeda with all PAPER_273276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g,\text{sum}}^{(26)} + \Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at t=0 (current epoch):

Term	Value (m/s)	Paper
g_grav = \mu_s\nabla(M_s/r)	1.227\times10?	baseline

Ug_sum (26 layers)	6.380\times10??	26-layer Triadic
?-term	2.998\times10?6	cosmological
g_quantum	~2\times10?6	HUP
g_Lorentz	~2\times10?6	IGM EM
g_fluid	~8\times10?	IGM buoyancy
F_res(?_HI) at t=0	1\times10?	PAPER_274
g_DM	1.356\times10?	PAPER_275
g_expansion at t=0	0	PAPER_276
a_dust	4.29\times10??	PAPER_276
?_approach	1.001001	PAPER_273

Note: g_expansion grows linearly with t, dominating over g_grav at t = t_H.

---

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`:
1. `H0` = 70.0 km/s/Mpc
2. `Omega_m` = 0.3
3. `Omega_Lam` = 0.7
4. `Mpc_to_m` = 3.086\times10
5. `rho_dust` = 1\times10? kg/m
6. `M_visible` = (1f_DM)M
7. `M_DM_mass` = f_DMM

---

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

1. **Friedmann-UQFF Expansion Coupling**: g_expansion = g_base – H(z) t. At t = t_H, adds 98.7% of g_base near-gravitational doubling over cosmological time.

2. **H_UQFF = 0.987**: New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship H0 t_H 1 in flat ?CDM. Andromeda's blueshift (z = -0.001) makes H_UQFF 0.15% below flat-universe value a blueshift Friedmann suppression.

3. **M_visible/M_DM_mass cascade + dust drag**: Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure (a_dust 4\times10?? m/s).

.Groups[1].Value

H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g_expansion = (μ_s∇(M_s/r)) H(z) t. For Andromeda's blueshift redshift z
= -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = 2.269$\times$10?8 s-1. Evaluated at the Hubble timescale t = t_H = 4.352$\times$10-7 s, the dimensionless Friedmann-UQFF Resonance Coefficient H_UQFF =
H(z)$\cdot$t_H **0.987** a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M_visible/M_DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

---

1. Background

The UQFF framework computes total gravitational acceleration g_total(r,t) as a sum of DPM-seeded, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER_273275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)t 1 implying that expansion-mediated gravitational coupling is of the same order as the Step 10 Newton observational projection term.

---

2. Friedmann H(z) Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_Lambda}$$

For Andromeda parameters: H0 = 70.0 km/s/Mpc, O_m = 0.3, O_? = 0.7, z = -0.001.

$$H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: H_SI = 69.969 $\times$ 10 / 3.086$\times$10 = **2.269$\times$10?8 s-1**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda:
- g_base = μ_s∇(M_s/r) = 6.674$\times$10? 1.989$\times$104 / (1.04$\times$10) = **1.227$\times$10? m/s**
- At t = 1 Gyr (3.156$\times$10-6 s): g_expansion = 1.227$\times$10? 2.269$\times$10?8 $\times$ 3.156$\times$10-6 = **8.79$\times$10? m/s**
- At t = t_Hubble (4.352$\times$10-7 s): g_expansion = 1.227$\times$10? $\approx$ 0.987 = **1.211$\times$10? m/s**

---

3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where t_H = 4.352$\times$10-7 s (Hubble time 13.8 Gyr).

For Andromeda (z = -0.001):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

H_UQFF $\approx$ 0.987 **1** is a remarkable result. It means:

> Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the Step 10 Newton observational projection term** nearly doubling g_base.

This near-unity value is not coincidental. In a flat ?CDM universe (O_m + O_? = 1):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because H0 t_H is the dimensionless Hubble number $\approx$ 0.96§1.0 for observationally consistent H0.

New UQFF Constant: H_UQFF = H(z)$\cdot$t_H **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269$\times$10?8 s-1
t_H	4.352$\times$10-7 s
H_UQFF	**0.987** (~1)
g_expansion(t_H)	1.211$\times$10? m/s
g_base	1.227$\times$10? m/s
Ratio g_exp/g_base	98.7%

3.3 Blueshift Sensitivity

For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O_m(1+z) is suppressed below O_m for z < 0), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

For Andromeda (z = -0.001): H_UQFF = 0.987 vs flat-universe H_UQFF0 = 0.9985. The blueshift suppresses H_UQFF by 0.15%.

This is the inverse of the redshift behaviour (z > 0 ? H(z) > H0 for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

---

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where ?_mean = M/V_fluid = 1.989$\times$104 / 106 = 1.989$\times$10?8 kg/m.

Numerical value for Andromeda:
$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

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---

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The full UQFF 2.0 equation for Andromeda with all PAPER_273276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g,\text{sum}}^{(26)} + \Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at t=0 (current epoch):

Term	Value (m/s)	Paper
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Note: g_expansion grows linearly with t, dominating over g_grav at t = t_H.

---

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8. Conclusions

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3. **M_visible/M_DM_mass cascade + dust drag**: Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure (a_dust 4$\times$10?? m/s).

UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]?r/GM = 5.7e-1§5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s at r_ISCO.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

<!-- PKG-DM-S225 -->

Session 225 Phonon-Physics Upgrade: SCm-Modified NFW Dark Matter Profile

> *Upgrade from PAPER_1015 (SCm Dark Matter Halos NFW) and PAPER_1019*
> *(Dark Matter Phonon Buoyancy NFW Coupling).*

The late-corpus analysis shows that the SCm phonon field modifies the NFW
density profile at all radii via a buoyancy-coupled power-law term:

$$\rho_{\text{UQFF}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)\left(1+\frac{r}{r_s}\right)^2} \times \left[1 + H_{\text{SCm}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{r_s}{r}\right)^{\alpha_{\text{phonon}}}\right]$$

where:
- $\alpha_{\text{phonon}} = 0.3$ governs the radial decay of phonon coupling

- $\beta_i = 0.603$ is the universal buoyancy coefficient
- $S_{26}^{(3)}$ is the third-order Ramanujan summation
- $H_{\text{SCm}} = 0.99$ is the manifold completeness factor

Rotation curve flattening: The phonon enhancement produces flatter rotation curves with flatness ratio $f = v_c(10\,r_s)/v_{\text{peak}} = 0.891$, compared to pure NFW $f \approx 0.75$. Peak circular velocity $v_{\text{peak}} \approx 204\;\text{km/s}$ for $M_{\text{halo}} = 10^{12}\,M_\odot$, $c = 10$.

Halo stabilization: The effective buoyancy pressure $P_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot \beta_i$ prevents cusp-core divergence, providing a physical mechanism for observed cored profiles without invoking SIDM cross-sections.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

---

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.064 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 17, \quad n_{\mathrm{channel}} = 17/26$$

Since $p_{\mathrm{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.064	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 17$	PASS Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

---

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1$\times$10-52 m-2 (UQFF vacuum term)	1.114$\times$10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day $\to$ Γ_p suppression	< 4.17$\times$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes
significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF–SM
bridge.*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1015	SCm Dark Matter Halos NFW Rotation Curve
PAPER_1019	Dark Matter Phonon Buoyancy NFW Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed

11 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{neutron}^{SCm} = N_n \, \sigma_n^{SCm}(\omega) \, \Phi_{phonon} \, (F_{U,Bi}/F_U - 1)$
where $\sigma_n^{SCm}(\omega,n) = \sigma_0 \exp[-(\omega-\omega_{SCm})^2/(2\Gamma^2)] \, (1 + [SSq] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) \cdot Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n \, (1103+26390n) \, W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa \cdot i \cdot n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate

beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

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